

AI-Assisted Threat Analysis Log

Segment 1 — traffic_segment_1.csv

1. AI Tool Used

Tool: ChatGPT

ChatGPT was used as an AI-assisted security analysis tool to review the PCAP-derived CSV dataset and surface potentially suspicious patterns — communication relationships, protocol usage, and unusual traffic behaviour worth a closer look.

2. Dataset Information

Dataset analyzed: traffic_segment_1.csv

Source: Extracted from 2021-08-16-formbook.pcap

Records analyzed: 1,200 packets

The analysis set out to identify:

- Frequently communicating IP addresses
- Repeated communication patterns
- Potential automated behaviour
- Unusual external communication
- Port and protocol observations
- Large data transfer patterns

3. AI Prompt Used

The following prompt was submitted to ChatGPT to begin the traffic investigation:

```
You are assisting with network traffic analysis as a cybersecurity analyst.
```

```
I have provided a CSV file extracted from a PCAP network capture.  
Analyze the dataset and help identify potential security concerns.
```

```
Perform an initial traffic investigation focusing on:
```

1. Identify the most frequently communicating source and destination IP addresses.
2. Detect repeated communication patterns that may indicate automated activity or beaconing behavior.
3. Highlight unusual external IP addresses or unexpected communication relationships.
4. Analyze commonly used and unusual ports/protocols observed in the traffic.
5. Identify any large or repeated data transfer patterns that require

investigation.

6. Point out any traffic patterns that could represent suspicious behaviour.

For every observation:

- Explain why it may be interesting from a security perspective.
- Mention the relevant evidence from the dataset.
- Clearly separate confirmed observations from assumptions.
- Suggest additional validation steps a security analyst should perform.

Do not label traffic as malicious without sufficient evidence.

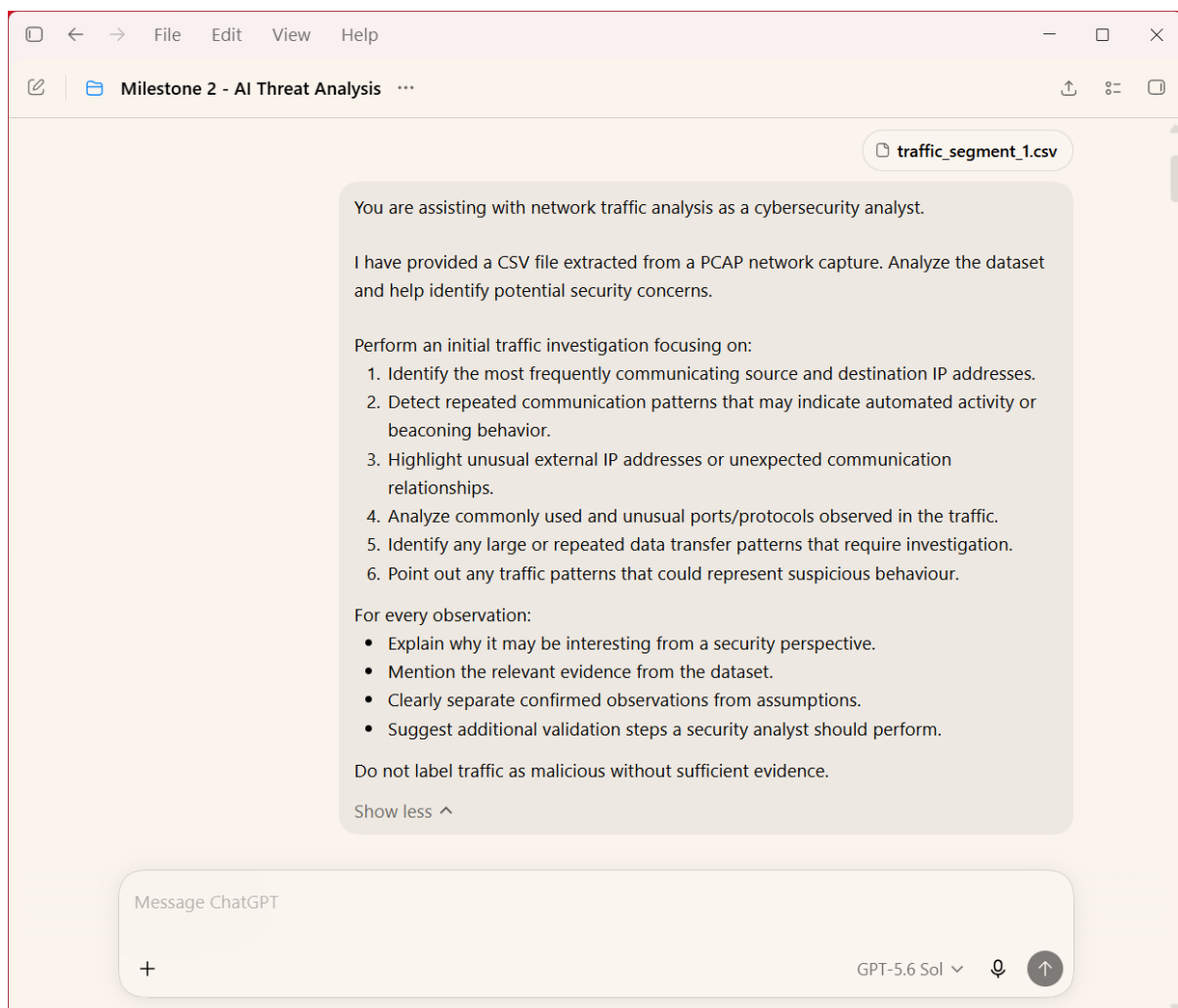


Figure 1: AI Prompt Submitted for Segment 1 Analysis

4. AI Generated Initial Findings (Segment 1)

Initial Network Traffic Investigation Report

Dataset: traffic_segment_1.csv

Records analyzed: 1,200 packets

Analysis goal: Identify potentially interesting traffic patterns from a PCAP-derived CSV dataset

The findings below are traffic observations only. An unusual pattern doesn't automatically mean malicious activity, and each item here requires additional validation before it can be classified as a security incident.

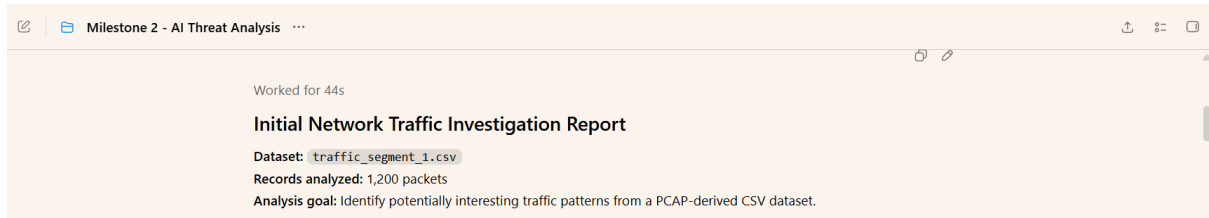


Figure 2: AI Generated Initial Analysis Response

Screenshots/05_AI_Initial_Analysis_Response.png

4.1 Most Frequently Communicating Source and Destination IP Addresses

Internal host 10.10.0.33 is the dominant communicating endpoint

Source IP	Packet Count
10.10.0.33	583
18.184.26.60	300
88.221.0.167	212
72.21.81.200	35
13.107.42.23	28

Destination IP	Packet Count
10.10.0.33	612
18.184.26.60	290
88.221.0.167	182
72.21.81.200	37
13.107.42.23	17

10.10.0.33 is both the busiest source and the busiest destination in the capture, accounting for roughly half of all traffic — 583 packets as a source and 612 as a destination, well ahead of the next most active address, 18.184.26.60. A host generating this much traffic isn't unusual on its own; it could just be an active user workstation. But the volume alone makes it the natural starting point for closer inspection, since it communicated with several distinct external addresses over the capture window.

Whether 10.10.0.33 is a user endpoint browsing the internet or a host running some kind of automated application can't be determined from packet counts alone. Confirming that would mean checking who owns the asset, what OS and software it runs, what the user was doing during the capture window, what DNS queries it generated, and whether EDR or antivirus logs show anything relevant.

4.2 Repeated Communication Patterns and Possible Automated Activity

High-volume repeated communication with 18.184.26.60

```
18.184.26.60 -> 10.10.0.33
Packets: 300
Bytes: 429,008

10.10.0.33 -> 18.184.26.60
Packets: 290
Bytes: 19,197

Median packet interval (10.10.0.33 -> 18.184.26.60): ~0.000036 seconds
```

The exchange between these two hosts is heavily asymmetric — 18.184.26.60 sends over twenty times as much data as it receives, and the 290 outbound packets from 10.10.0.33 are spaced roughly 36 microseconds apart on average, fast enough to suggest an automated exchange rather than manual browsing.

That kind of pattern shows up in plenty of legitimate contexts — web browsing, software updates, application synchronization, cloud services — but it's also consistent with automated polling or command-and-control beaconing. Nothing here confirms malware; there's no payload inspection behind this observation. It's worth pulling the TCP stream contents, checking the TLS certificate, identifying the process responsible for the connection, and reviewing DNS history and application logs.

4.3 Unusual External IP Addresses and Communication Relationships

The following external IPs communicated with internal host 10.10.0.33 during the capture:

External IP	Packets	Total Bytes	Notes
18.184.26.60	590	448 KB	Highest volume
88.221.0.167	394	300 KB	High repeated communication
72.21.81.200	72	37 KB	Moderate traffic
13.107.42.23	45	33 KB	Likely Microsoft-associated range
52.109.76.30	25	19 KB	Likely Microsoft-associated range
8.8.8.8	18	1.8 KB	DNS traffic

88.221.0.167 shows repeated large responses

```
88.221.0.167 -> 10.10.0.33
Packets: 212
Bytes: 286,470
Average packet size: ~1351 bytes
```

A transfer of this size from an external host is worth reviewing — it could be normal web content delivery, a software download, or CDN traffic, but it could equally be an unexpected download or data delivery from infrastructure that hasn't been identified yet. Without domain information, application context, or payload analysis, there's no way to tell which. A WHOIS lookup, reverse DNS, reputation check, TLS certificate review, and inspection of the HTTP Host header would help narrow it down.

4.4 Ports and Protocol Analysis

TCP destination ports observed

Port	Count	Likely Service
80	300	HTTP
443	253	HTTPS
53480	260	Dynamic/private application port
53464–53479	Multiple	Ephemeral/application ports

UDP destination ports observed

Port	Count	Possible Service
53	9	DNS
137	13	NetBIOS Name Service
1900	8	SSDP/UPnP

Protocol findings

Protocol	Packets
TCP	1044
TLS	85
DNS	18
NBNS	13
SSDP	8

Some of the captured traffic used plain HTTP on port 80, which doesn't encrypt its payload. It's worth identifying which domains were accessed over HTTP and whether any sensitive information passed in the clear.

SSDP discovery traffic also appears in the capture, addressed to 239.255.255.250 over UDP port 1900. SSDP is a normal part of UPnP device discovery, though UPnP-enabled devices can end up exposing services they don't need to. The capture confirms SSDP discovery traffic is present; there's no evidence of any exploitation attempt against it.

4.5 Large or Repeated Data Transfer Patterns

The largest data transfers observed in the capture:

Source	Destination	Bytes
18.184.26.60	10.10.0.33	429 KB
88.221.0.167	10.10.0.33	286 KB
72.21.81.200	10.10.0.33	33 KB
13.107.42.23	10.10.0.33	31 KB

Several external-to-internal transfers here are larger than a typical single-packet exchange. That's not unusual by itself — most large transfers are web downloads, application content, or updates — but it's worth a closer look if the user didn't initiate the activity, the destination isn't recognized, or the same transfer repeats. Nothing in the dataset points to data exfiltration, a malware download, or an unauthorized transfer, though none of those can be ruled out from metadata alone.

4.6 Potentially Suspicious Traffic Patterns

Finding 1 — Automated-looking repeated communication

Hundreds of packets were exchanged with both 18.184.26.60 and 88.221.0.167, and the frequency and timing of that exchange suggest automated application behaviour rather than manual use. The pattern is real, but nothing confirms it's malicious — confidence: medium.

Finding 2 — Multiple external services communicating with one internal host

10.10.0.33 communicated with five external addresses over the capture: 18.184.26.60, 88.221.0.167, 72.21.81.200, 13.107.42.23, and 52.109.76.30. A compromised endpoint could look like this, but so could any modern system that routinely reaches out to multiple external services — confidence: low.

Finding 3 — Presence of NetBIOS and SSDP traffic

UDP 137 (NetBIOS) and UDP 1900 (SSDP) traffic both appear in the capture. Neither is inherently malicious, but both are protocols that are often unnecessary in a hardened environment — confidence: low.

Overall Assessment

Across the capture, 10.10.0.33 stands out as the main communicating host, with 18.184.26.60 and 88.221.0.167 accounting for the highest-volume external communication. Traffic is dominated by HTTP, HTTPS/TLS, DNS, NBNS, and SSDP, and includes several large inbound transfers from external systems.

Concern	Confidence
Automated communication pattern	Medium
Unknown external service relationships	Medium
SSDP/NBNS exposure	Low
Malicious activity	Not confirmed

Recommended Next Investigation Steps

1. DNS analysis — extract requested domains and confirm whether the external IPs map to legitimate services.
2. TCP stream reconstruction — follow the conversations with 18.184.26.60 and 88.221.0.167.
3. Threat intelligence lookup — check reputation of the external IPs.
4. TLS inspection — review certificate issuer, subject name, and validity.
5. Endpoint investigation — identify the process generating traffic from 10.10.0.33.
6. Baseline comparison — determine whether this traffic is normal for the device.

Final Conclusion

The dataset shows high-volume, repeated communication between internal host 10.10.0.33 and several external addresses, most notably 18.184.26.60 and 88.221.0.167. The frequency and volume make these worth investigating further, but packet metadata alone isn't enough to call the traffic malicious. That determination needs a look at packet contents, DNS information, TLS metadata, and endpoint activity.

5. Analyst Review of AI Output

The AI-generated findings were checked against the extracted CSV dataset directly.

Correct AI Observations

The AI got the following right:

Internal host identification

It classified internal and external addresses using the standard RFC 1918 private ranges — 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16 — treating anything within those ranges as internal and anything outside them as external.

External communication identification

It correctly flagged 18.184.26.60, 88.221.0.167, 13.107.42.23, and 52.109.76.30 as external endpoints, all of which fall outside the private IP ranges and are publicly routable.

Traffic pattern identification

It also picked out the high-volume communication, the multiple external relationships, the TCP/TLS/DNS traffic, and the repeated packet exchanges — all consistent with what's actually in the packet metadata.

6. Limitations of AI Analysis

The AI's findings are useful starting points, but they can't independently confirm malicious activity — the analysis was based on packet metadata alone. Confirming or ruling out a security incident would still require:

- Packet payload inspection
- DNS analysis
- TLS certificate inspection
- Threat intelligence checks
- Endpoint investigation

7. AI Accuracy Assessment

Category	Assessment
IP classification	Correct
Protocol identification	Correct
Communication pattern detection	Useful
Malware determination	Not confirmed
Threat attribution	Not possible from metadata alone

Segment 1 Summary

Category	Observation
Primary internal host	10.10.0.33
Highest external communication	18.184.26.60
Main protocols	TCP, TLS, DNS
Suspicious pattern	Repeated communication
Risk level	Requires investigation
Confirmed malware	No

8. Conclusion

ChatGPT was used here to generate initial hypotheses from the network traffic dataset, and it did a solid job flagging the areas that warranted a closer look. Every finding was still treated as an indicator rather than a confirmed threat, and each one was checked against the dataset before being accepted.